IndianOil

# JOURNEY RISK MANAGEMENT (JRM) STUDY

## Salem Terminal to THULASIMANI AGENCIES

### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

## Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

## Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## Environmental Considerations

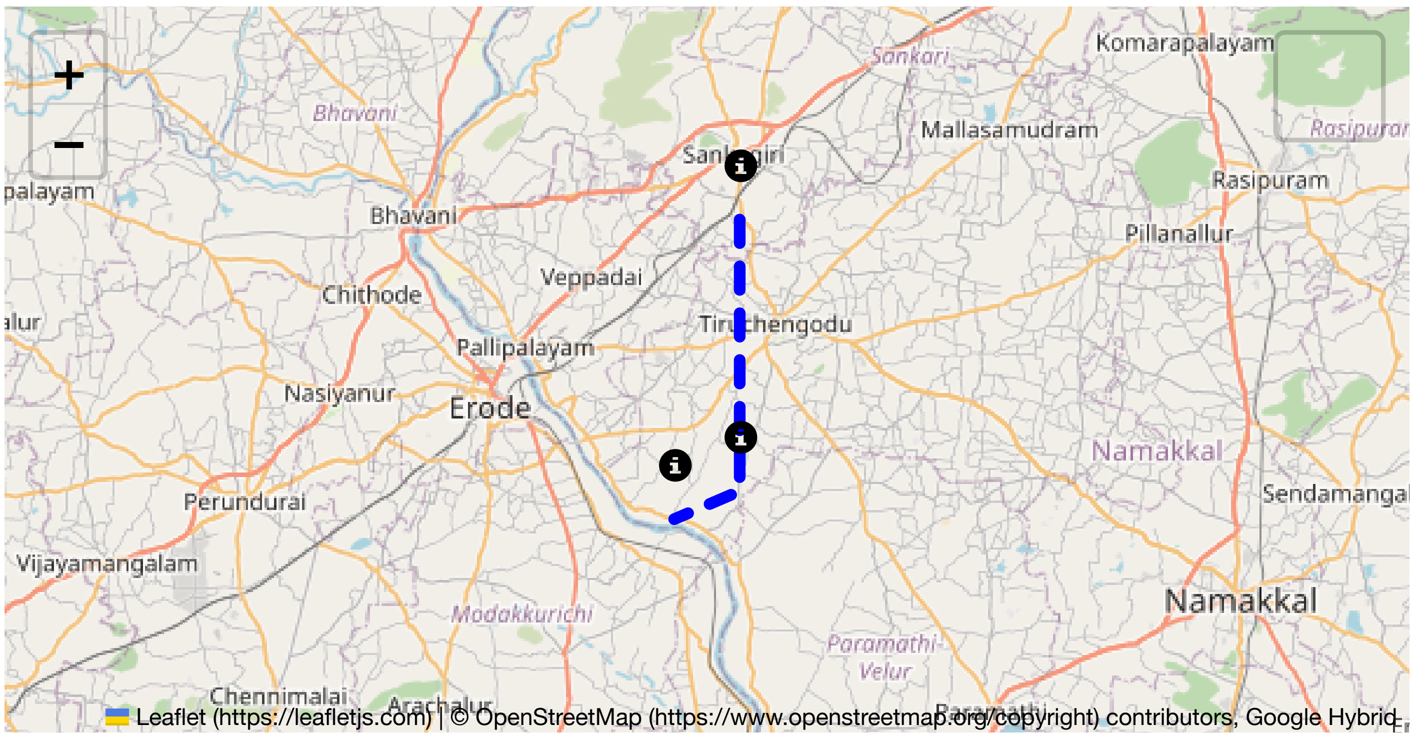
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

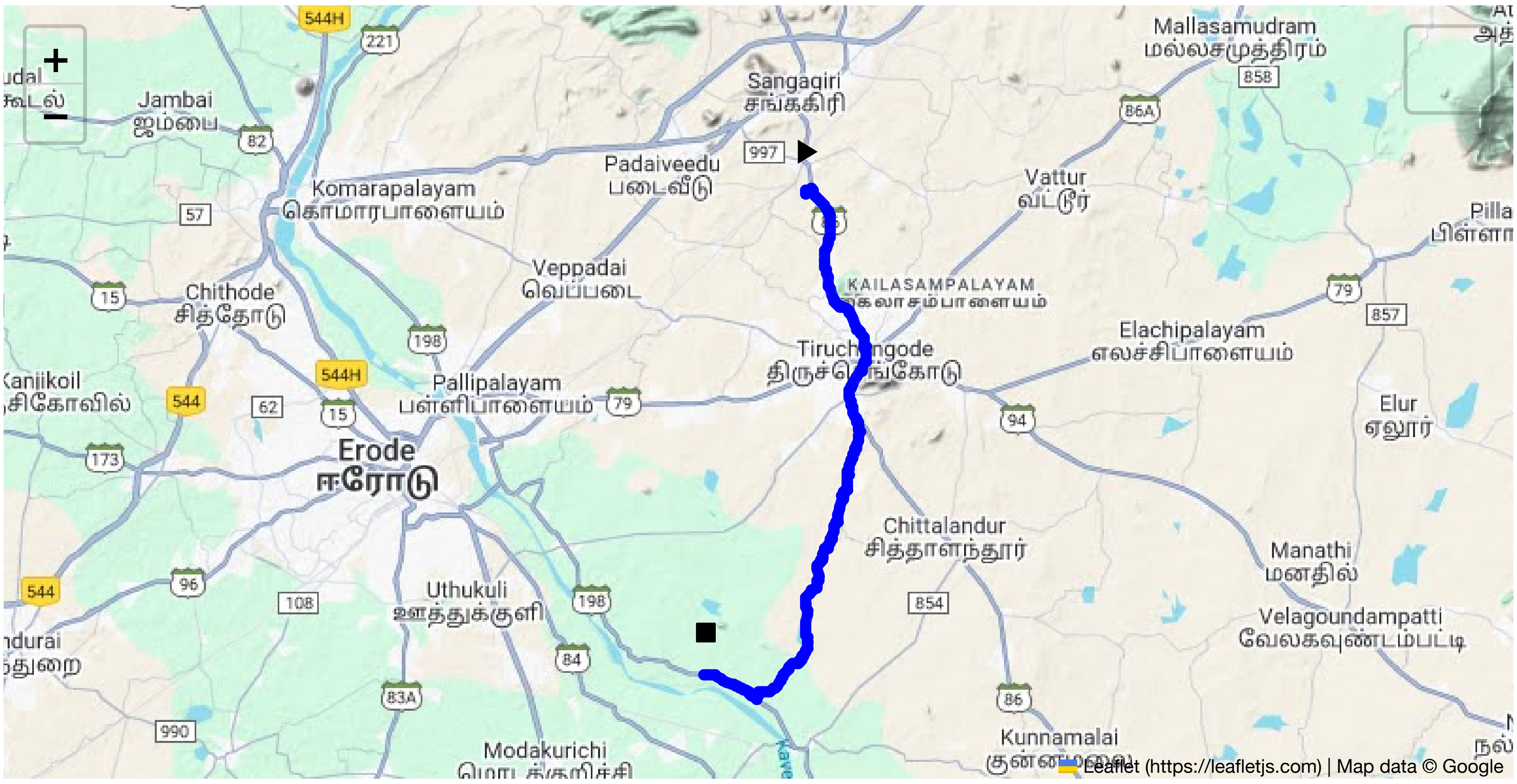
## Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.




**Route Summary:**
**Total Distance: 25.97 km**
**Estimated Duration: 0.7 hours**
**Adjusted Duration (Heavy Vehicle): 0.9 hours**

**Start: (11.4381, 77.8734)**
**End: (11.266364, 77.836037)**




## Welcome to the Journey Risk Management Study

## 1. Overview of the Route Map

The route from Sangagiri to Iraiyamankalam via PMG Garden in Aandivalasu is approximately 25.97 kilometers. This route traverses rural and semi-urban regions, primarily through local and state highways. Drivers will pass through smaller towns and agricultural areas, suggesting varying traffic and road conditions.

## 2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences a tropical climate with hot summers, heavy monsoons (from June to September), and mild winters. During monsoon season, the routes may face issues like waterlogging, especially in low-lying areas, potentially making driving hazardous due to slippery roads and reduced visibility. Additionally, occasional cyclones may affect this region, bringing intense rainfall and strong winds, thus delaying travel or requiring rerouting.

## 3. Analysis of Traffic Patterns

Traffic peaks in the mornings (7:00-9:00 AM) and evenings (5:00-7:00 PM) with typical congestion near market areas and smaller urban centers. Local events or festivals can also cause spikes in traffic. While major state highways experience regular traffic flow, narrower roads in towns can slow down even further during peak times.

## 4. Assessment of Road Quality and Infrastructure

The route includes a mix of well-maintained state highways and narrower country roads. While major roads generally offer good quality, secondary roads might have untreated potholes, worn-out markings, minimal signage, and poorer lighting conditions at night. Construction or maintenance work may also occur periodically, requiring detours or careful navigation.

## 5. Suggestions for Alternative Routes for Emergencies

While specific shortcuts may not be advisable due to narrow paths or lesser-known routes not suitable for heavy vehicles, the closest alternative during emergencies would be to use parallel state highways that run east-west just north of the primary route; however, these must be checked for their capacity to handle heavy vehicles and hazardous materials.

## 6. Summary of Local Regulations Affecting Hazardous Material Transport

Drivers must comply with local regulations for transporting hazardous materials, requiring permits and strict adherence to prescribed routes if specified. Transporting during night hours might be restricted or limited. Ensure all documentation is up-to-date, and emergency contact numbers are available for quick access.

## 7. Overview of Historical Incidents

Recent historical data does not suggest frequent incidents involving heavy vehicles or hazardous materials on this specific route. However, general caution is advised due to occasional reports of accidents in the broader region caused by factors like vehicle malfunction or human error.

## 8. Environmental Considerations and Sensitive Areas

The route crosses near agricultural zones, where accidental spills could have serious environmental impacts. Drivers should be aware of local wildlife, particularly in more rural segments. It’s crucial to ensure no contamination occurs, vent premature container leaks and adhere to set routes avoiding reserve or protected areas.

## 9. Analysis of Communication Coverage

Generally, network coverage is stable in towns and along major roads, but connectivity may drop in more remote agricultural areas or near hilly terrain. Drivers should prepare for possible dead zones by ensuring all communication devices are fully charged and carrying emergency contact methods like satellite phones.

## 10. Estimated Emergency Response Times

Urban centers along the route can expect quicker emergency response times of roughly 15-20 minutes. However, in rural stretches, response times might extend to 30-45 minutes, depending on the availability and proximity of emergency services.

## 12. Overall Summary of Risk Assessment

The route presents moderate risk mainly due to local weather conditions, surrounding environments, and road quality variances. Planning for unpredictable weather, enhancing route understanding, preparing for network issues, and ensuring compliance with local regulations are critical for safe travel. Utilizing advanced route navigation tools and maintaining regular communication with dispatch can further mitigate risks for drivers transporting hazardous materials along this route.

## Risk Assessment - Turns

| | Risk Type | Risk Level | Coordinates | Speed Limit |
|---|---|---|---|---|
| 0 | Roundabout | High | 11.38395, 77.89480 | 15 KM/Hr |
| 1 | Turn | High | 11.43811, 77.87348 | 15 KM/Hr |
| 2 | Turn | Medium | 11.43956, 77.87340 | 30 KM/Hr |
| 3 | Turn | High | 11.43968, 77.87345 | 15 KM/Hr |
| 4 | Blind Spot | Blind Spot | 11.44029, 77.87544 | 10 KM/Hr |
| 5 | Turn | High | 11.37868, 77.89498 | 15 KM/Hr |
| 6 | Turn | High | 11.37850, 77.89453 | 15 KM/Hr |
| 7 | Turn | Medium | 11.35759, 77.89191 | 30 KM/Hr |
| 8 | Turn | Medium | 11.33071, 77.88660 | 30 KM/Hr |
| 9 | Turn | High | 11.33065, 77.88650 | 15 KM/Hr |
| 10 | Turn | Medium | 11.32134, 77.88489 | 30 KM/Hr |
| 11 | Turn | Medium | 11.31205, 77.88131 | 30 KM/Hr |
| 12 | Turn | Medium | 11.31151, 77.88008 | 30 KM/Hr |
| 13 | Turn | Medium | 11.30805, 77.87934 | 30 KM/Hr |
| 14 | Turn | High | 11.29763, 77.87710 | 15 KM/Hr |
| 15 | Turn | High | 11.29668, 77.87515 | 15 KM/Hr |
| 16 | Turn | Medium | 11.29620, 77.87503 | 30 KM/Hr |
| 17 | Turn | High | 11.29585, 77.87457 | 15 KM/Hr |
| 18 | Turn | High | 11.29549, 77.87464 | 15 KM/Hr |
| 19 | Turn | Medium | 11.27061, 77.86833 | 30 KM/Hr |
| 20 | Turn | High | 11.26017, 77.85606 | 15 KM/Hr |
| 21 | Blind Spot | Blind Spot | 11.25802, 77.85511 | 10 KM/Hr |

## Emergency Locations

| | type | name | coordinates | speed_limit | risk_level |
|---|---|---|---|---|---|
| 0 | hospital | Tiruchengode, Goverment Hospital | 11.3903328, 77.8920627 | 30 km/h | Medium |
| 1 | hospital | SPM Medical Centre,Tiruchengode | 11.3881331, 77.8931963 | 30 km/h | Medium |
| 5 | hospital | T.C.A Hospital Tiruchengode | 11.3791885, 77.8965774 | 30 km/h | Medium |
| 6 | hospital | Soorya Multispecialty Hospital | 11.3786429, 77.8931912 | 30 km/h | Medium |
| 7 | clinic | Kongu Nursing Home | 11.3783065, 77.8961134 | 30 km/h | Medium |

| | type | name | coordinates | speed_limit | risk_level |
|---|---|---|---|---|---|
| 8 | hospital | Tiruchengode Government Hospital | 11.37645, 77.89426 | 30 km/h | Medium |
| 9 | hospital | Krishna Hospital, Namakkal | 11.3754811, 77.8931817 | 30 km/h | Medium |
| 10 | hospital | Tirukumaran Hospitals | 11.3782118, 77.8914933 | 30 km/h | Medium |
| 13 | police | கரட்டுப்பாளையம் காவல் நிலையம் | 11.357586, 77.8922539 | 30 km/h | Medium |

## Crowded Spots

| | type | name | coordinates | speed_limit | risk_level |
|---|---|---|---|---|---|
| 2 | school | அரசு ஆண்கள் மேல்நிலைப் பள்ளி | 11.3850439, 77.8948279 | 30 km/h | Medium |
| 3 | school | அரசு பெண்கள் மேல்நிலைப் பள்ளி | 11.3844247, 77.8949053 | 30 km/h | Medium |
| 4 | marketplace | திருச்செங்கோடு தினசரி காய்கறி சந்தை | 11.3833608, 77.8970145 | 30 km/h | Medium |
| 11 | school | KSR Educational institution | 11.3772013, 77.8908807 | 30 km/h | Medium |
| 12 | school | MDV School | 11.3719843, 77.8915244 | 30 km/h | Medium |

## Route Photos of Risky Spots


**Risk Type:** Roundabout

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.38395, 77.89480


**Risk Type:** Blind Spot

**Risk Level:** Blind Spot

**Speed Limit:** 10 KM/Hr

**Coordinates:** 11.44029, 77.87544


**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.37868, 77.89498


**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.37850, 77.89453


**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.35759, 77.89191


**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.33071, 77.88660


**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.33065, 77.88650


**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.32134, 77.88489


**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.31205, 77.88131


**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.31151, 77.88008


**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.30805, 77.87934


**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.29763, 77.87710


**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.27061, 77.86833


**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.26017, 77.85606


**Risk Type:** Blind Spot

**Risk Level:** Blind Spot

**Speed Limit:** 10 KM/Hr

**Coordinates:** 11.25802, 77.85511